PhysWM: Grounding Predictive World Models in Physical Equations

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Abstract

Predictive world models can forecast dynamics without organizing their representations around the physical variables governing a system. We introduce PHYSWM, a two-path architecture that grounds an action-conditioned predictive world model through known physical equations. A neural path predicts the next state from data, while an equation path maps the same predictive latent through a low-capacity probe to physical variables and a frozen differentiable solver. Without parameter labels, the equation path learns to reproduce the neural path's stopped-gradient prediction. We evaluate grounding through prediction, semantic readout, parameter substitution, and multi-horizon equation rollouts. On PokeWorld, the predictor beats persistence and inferred variables reduce seven-step rollout error relative to shuffled variables across three seeds. On randomized PushT, inferred effective dynamics beat shuffled episode assignments in all three seeds. The results support equation-compatible effective coordinates while exposing limits from non-identifiability and simulator mismatch.

1 Introduction

World models compress observations and actions into representations useful for predicting future dynamics (4; 5; 6; 10). Predictive accuracy alone, however, does not ensure that these representations expose mass, damping, stiffness, friction, or control response. A high-capacity predictor may exploit trajectory correlations while leaving the governing physical factors entangled, even when its next-state predictions are accurate (9). This separates predicting *what happens* from representing *why the system evolves that way*.

We study whether known physical equations can ground a flexible predictive world model without physical-parameter annotations. PHYSWM produces two next-state predictions from the same action-conditioned latent (Figure 2). Path A is a learned decoder trained on observed transitions. Path B maps the latent to a compact vector $\hat{\theta}$ and evaluates a frozen differentiable solver. Path B targets Path A's stopped-gradient prediction rather than the dataset state directly. The neural path remains anchored to data and free to absorb model mismatch; the physical path asks whether the model's predictive belief admits a compact equation-based explanation. This fixed-decoder role is related to differentiable system identification (8), but our training target is the predictor's belief rather than rendered pixels or labeled physical parameters.

Our contribution is threefold. First, we introduce a shared-latent, two-path objective for grounding predictions in a supplied equation family without parameter labels. Second, we evaluate grounding functionally by changing only the solver's parameter vector while holding states, actions, weights, and equations fixed. Third, we distinguish *semantic parameters*, which match named ground-truth quantities, from *effective physical coordinates*, which improve equation rollouts but may compensate for non-identifiability or solver mismatch.

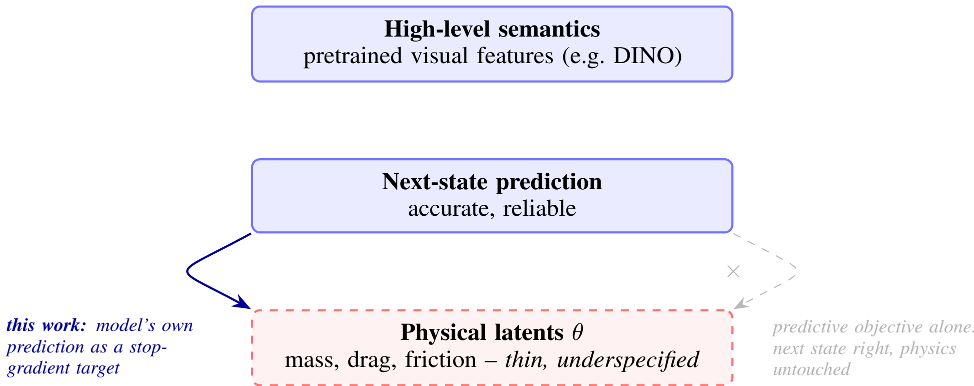

Figure 1: **Filling a thin foundation from the top.** Self-supervised encoders build a hierarchy that is tall at the top (rich semantics) but thin at the bottom (physical structure). A world model trained only for next-state prediction inherits this shape: it gets the top level right without the bottom level ever becoming physically meaningful. We use the reliable top-level signal, the model's own next-state prediction, as the training target for a low-capacity, physics-constrained probe at the bottom, without ever touching a physical-parameter label.

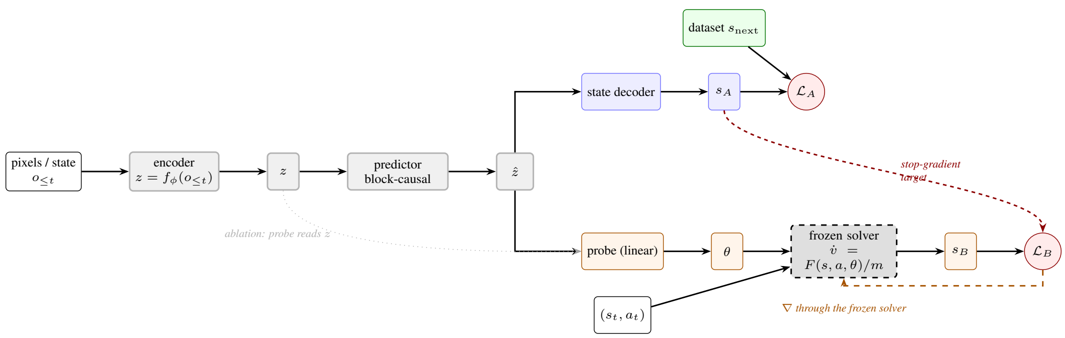

Figure 2: **PhysWM architecture.** Encoder and predictor form one shared trunk producing a single action-conditioned latent $\hat{z}$; both heads read $\hat{z}$, never a private view of the observation. Path A (blue) is the ordinary predictor: a decoder reads $\hat{z}$ to a state s_A, and $\mathcal{L}_A$ (Equation (4)) regresses the dataset's ground truth, exactly what a predictive world model already does. Path B (orange) reads a physics vector θ off the same $\hat{z}$ with a deliberately low-capacity probe, integrates it through a *frozen, zero-parameter* solver (gray, dashed) to get an independent estimate s_B. $\mathcal{L}_B$ does *not* use the dataset label: it regresses s_A with gradient stopped (red, dashed), so the only route back to a learnable parameter is through the fixed solver into θ (orange dashed arrow), and θ can only improve by becoming a better physical explanation of what Path A already predicts. The dotted arrow is not part of the method: it marks the pre-action routing ablation (Section 3), where the probe reads z instead of $\hat{z}$.

2 Equation-grounded predictive representations

Given observations o_t, actions a_t, and prediction state s_t, an encoder and causal action-conditioned predictor form

$$z_t = E_\phi(o_t), \qquad \hat{z}_t = P_\psi(z_{\le t}, a_{\le t}). \tag{1}$$

The neural decoder produces $\hat{s}^A_{t+1} = D_\omega(\hat{z}_t)$ and is trained in normalized state space:

$$\mathcal{L}_A = \|N(\hat{s}^A_{t+1}) - N(s_{t+1})\|_2^2. \tag{2}$$

A bounded low-capacity probe pools only causal context latents,

$$\hat{\theta} = \mathrm{Bound}(\rho_\xi(\mathrm{Pool}(\{\hat{z}_t\}_{t \in \mathcal{C}}))), \tag{3}$$

and a frozen solver S, containing no learnable parameters, produces $\hat{s}^B_{t+1} = S(s_t, a_t, \hat{\theta})$. The grounding loss is

$$\mathcal{L}_B = \|N(\hat{s}^B_{t+1}) - \mathrm{sg}(N(\hat{s}^A_{t+1}))\|_2^2, \qquad \mathcal{L} = \mathcal{L}_A + \alpha \mathcal{L}_B. \tag{4}$$

Table 1: Mean $\pm$ sample standard deviation over three seeds. RMSE is lower better; R^2 is higher better.

Benchmark	Metric	Baseline / control	PHYSWM
PokeWorld	Query prediction RMSE	Persistence 0.383 ± 0.052	0.282 ± 0.046
PokeWorld	Stiffness R^2	Predictive 0.754 ± 0.017	0.784 ± 0.029
PokeWorld	Mass R^2	Predictive 0.241 ± 0.028	0.237 ± 0.055
PokeWorld	Drag R^2	Predictive -0.191 ± 0.052	-0.146 ± 0.043
PokeWorld	7-step solver RMSE	Shuffled 0.220 ± 0.018	Inferred 0.193 ± 0.010
PushT	Solver substitution RMSE	Shuffled 0.267 ± 0.013	Inferred 0.208 ± 0.011

The stopped target prevents $\mathcal{L}_B$ from moving the Path-A decoder through its target edge, while gradients through S, ρ_ξ, and P_ψ shape the shared predictor. The default probe is linear, preventing it from hiding another world model. $\hat{\theta}$ is always a forward-pass output, never a free test-time optimization variable. The stop-gradient follows self-distillation methods (7; 1; 3); here it enforces a one-way dependency between a predictive teacher path and an equation-constrained student path.

Causal evaluation split. The identifier observes only completed context transitions. Metrics are computed on later query transitions, and no query outcome is accessible to the probe. With the default eight-frame window, four completed transitions form the context. Functional evaluation uses eight context transitions followed by seven held-out rollout transitions.

Grounding criterion. We call a coordinate physically grounded when reading it from the predictive latent allows the fixed equations to explain held-out dynamics better than a coordinate carrying no episode-specific information. This is deliberately weaker than claiming recovery of a unique material parameter: trajectories may identify only ratios such as stiffness/mass, and approximate solvers may induce compensating effective values.

3 Evaluation

We test four complementary properties. (1) *Predictive adequacy*: Path A must beat persistence, $\hat{s}_{t+1} = s_t$. (2) *Semantic readout*: where labels exist, a supervised linear probe measures held-out episode-level R^2. (3) *Parameter substitution*: the solver receives the inferred vector, another episode's shuffled vector, nominal constants, or true parameters when available. (4) *Multi-horizon rollout*: the same substitutions are rolled forward under fixed held-out actions. All errors are scaled per state dimension.

PokeWorld. This synthetic contact benchmark varies mass, contact stiffness, and drag while keeping object appearance fixed (9). Its differentiable solver matches the generator, providing true-parameter and solver-adequacy certificates. The reported runs use 2,048 episodes, a tiny CNN diagnostic encoder, and three seeds.

Randomized PushT. This planar manipulation benchmark extends the contact-rich Push-T task (2) with episode-level variation in effective dynamics. Its solver is approximate and ground-truth labels are unavailable for most coordinates, making inferred-versus-shuffled substitution the primary test. The reported runs use 512 episodes and three seeds.

4 Results and discussion

Accurate prediction is not uniform parameter recovery. On PokeWorld, Path A beats persistence in every seed (Table 1). Stiffness is strongly linearly readable and improves modestly under equation grounding. Mass remains weak, and drag has negative held-out R^2 for both models. Thus the results do not support uniform recovery of named parameters. They instead expose an observability boundary: an equation-compatible representation can be useful even when individual semantic factors are not uniquely recoverable.

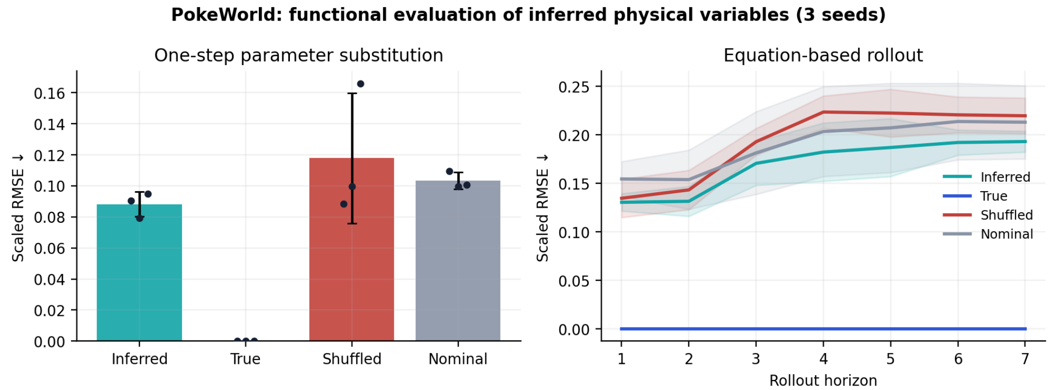

Figure 3: PokeWorld functional evaluation. Left: one-step error after substituting inferred, true, shuffled, or nominal parameters. Right: equation-rollout error over horizon. Curves and error bars show mean and sample standard deviation over three seeds.

Inferred coordinates improve equation rollouts. In PokeWorld one-step evaluation, inferred coordinates achieve 0.088 ± 0.008 scaled RMSE versus 0.118 ± 0.042 for shuffled and 0.103 ± 0.006 for nominal values. The one-step ordering is not uniform across seeds. At seven steps, however, inferred coordinates beat shuffled assignments in every seed, yielding 0.193 ± 0.010 versus 0.220 ± 0.018 (Fig. 3). True parameters are nearly exact because the solver matches the generator.

On PushT, inferred variables obtain 0.208 ± 0.011 substitution RMSE versus 0.267 ± 0.013 for shuffled assignments. Inferred values beat shuffled in all three seeds and nominal values in two of three. Solver sensitivity is concentrated in the agent's proportional and derivative gains, so this supports episode-specific effective control dynamics rather than complete contact-parameter identification. Moreover, Path A fails to beat persistence in one PushT seed, limiting the cross-benchmark claim.

Simulator mismatch changes the meaning of $\hat{\theta}$. A corrected FetchPush diagnostic illustrates the risk. After fixing a $10\times$ gripper-integration error, inferred variables still outperform true mass and friction by a large margin, while solver-space task success is identical for every parameter source. The probe is therefore compensating for residual equation mismatch. This motivates reporting solver adequacy and sensitivity alongside parameter accuracy and describing $\hat{\theta}$ as an effective physical coordinate unless semantic identifiability is independently certified.

5 Conclusion and limitations

PHYSWM grounds a predictive world model by requiring its shared action-conditioned representation to support a compact explanation through known physical equations. Current results show improved equation-based rollouts relative to shuffled episode variables, especially over longer horizons, but not uniform recovery of named parameters. The supplied equations, observation process, and solver fidelity determine what can be identified.

The study is limited to diagnostic tiny-CNN experiments on two usable multi-seed benchmarks. It does not discover equations, select informative actions, or demonstrate closed-loop planning or sim-to-real transfer. Corrected FetchPush and stable CartPole experiments remain incomplete, and pretrained visual encoders require paper-scale evaluation. These limitations define the next step: test whether equation grounding remains beneficial with strong visual backbones and whether uncertainty-aware actions can actively reveal otherwise ambiguous physical coordinates.

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